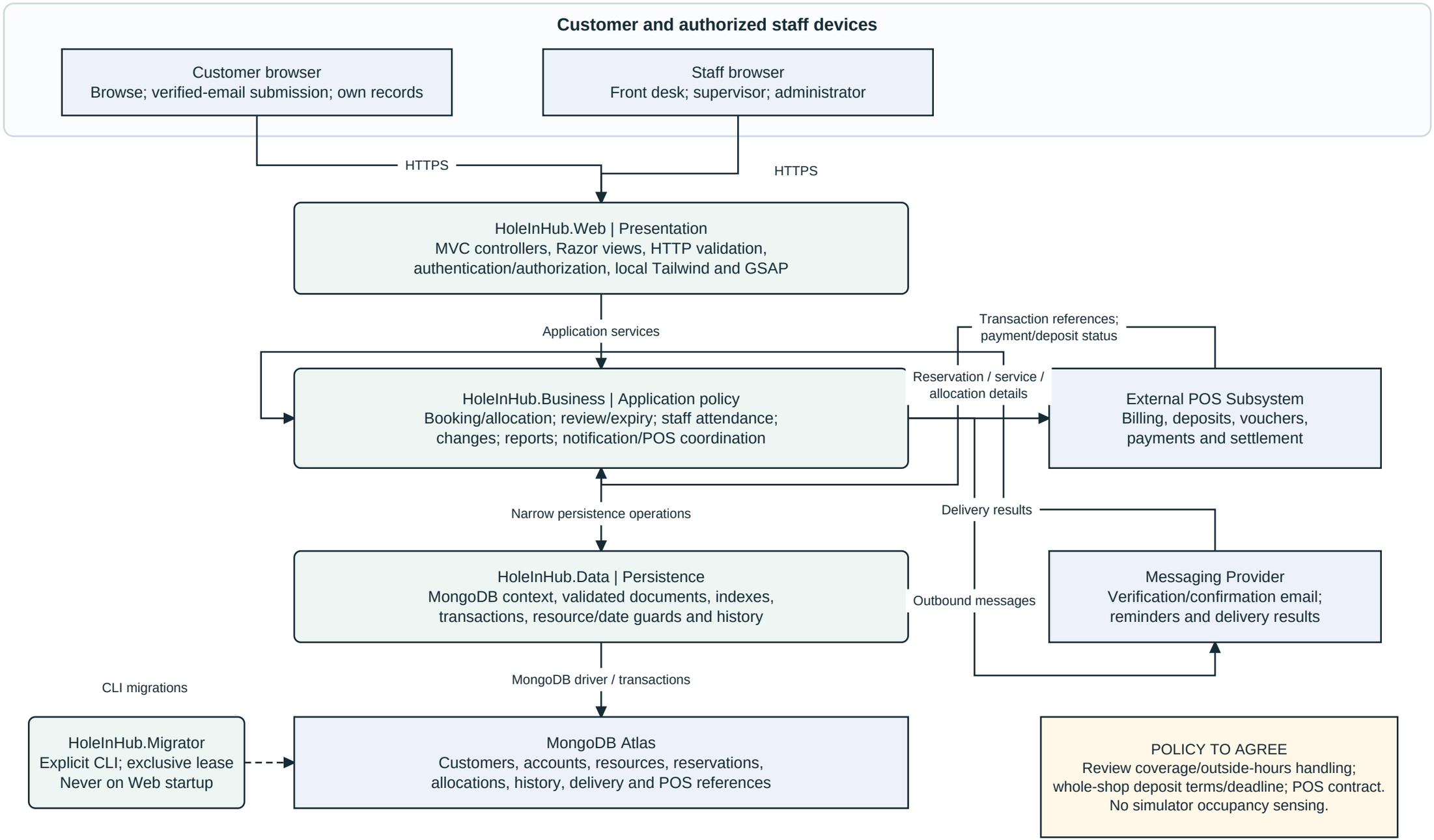
System Architecture Diagram

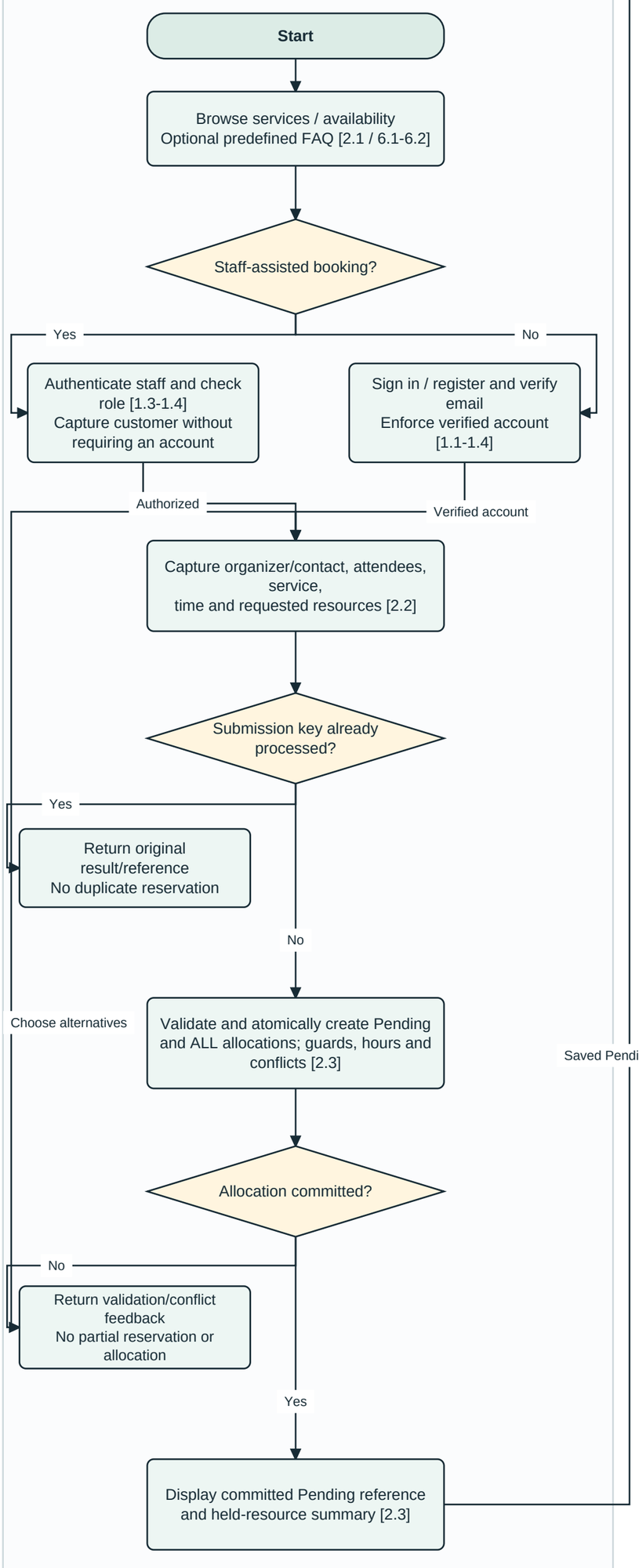
Target architecture | solid arrows: calls/data exchange; dashed arrow: explicit operational command



System Flowchart | Booking through Final Outcome

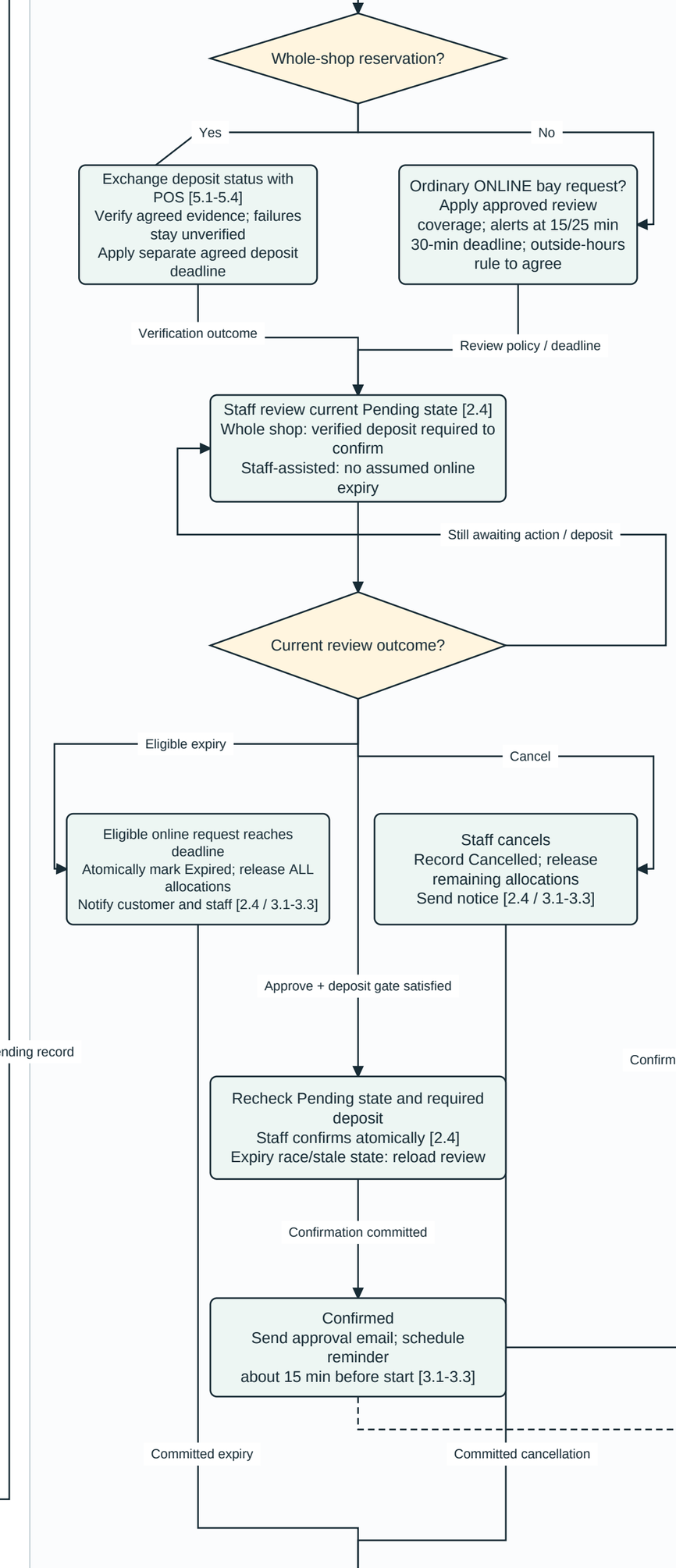
One connected workflow | numbered references map to the Level 2 DFD | diamonds are decisions

A. Access and atomic booking



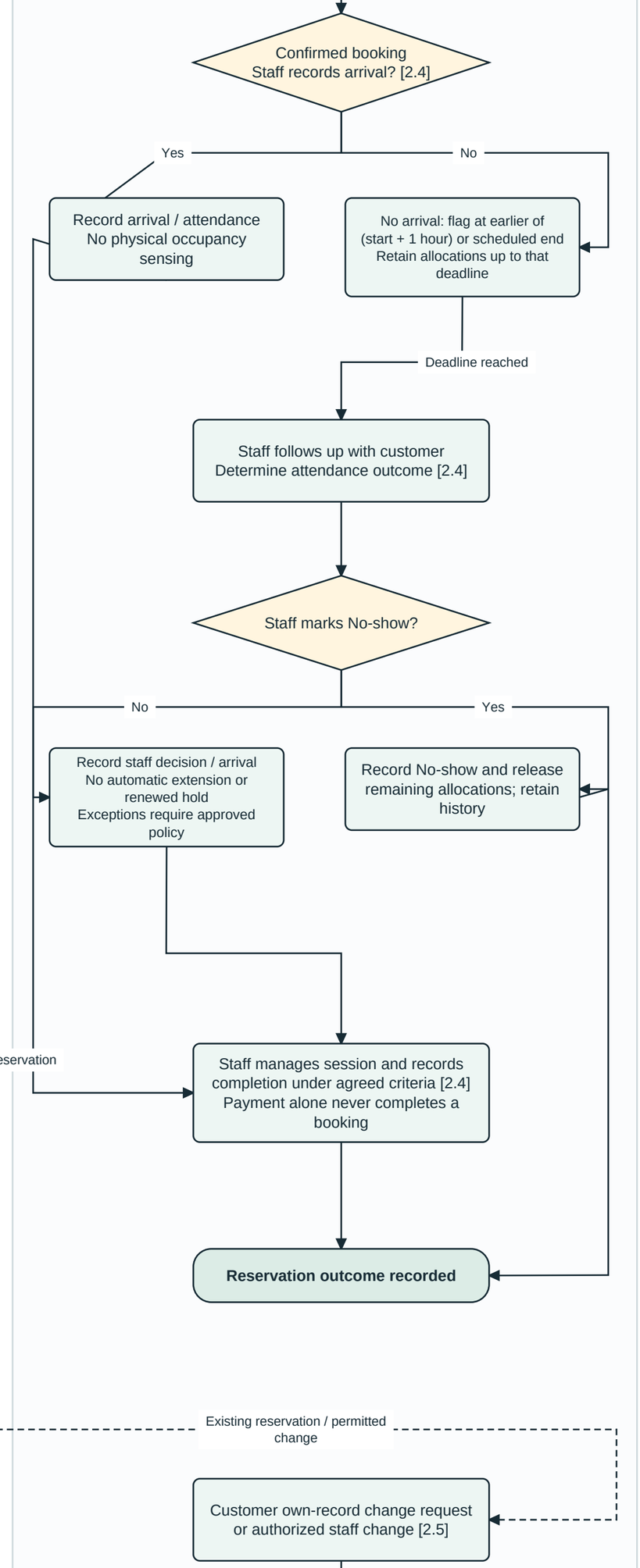
Authorized staff reports [4.1-4.3]
Counts, services, trends, cancellations/no-shows
and RESERVED-TIME utilization

B. Pending review and confirmation



CLIENT POLICY REQUIRED
Review hours and outside-hours clock; whole-shop deposit deadline/evidence; change/attendance exceptions and completion rules.

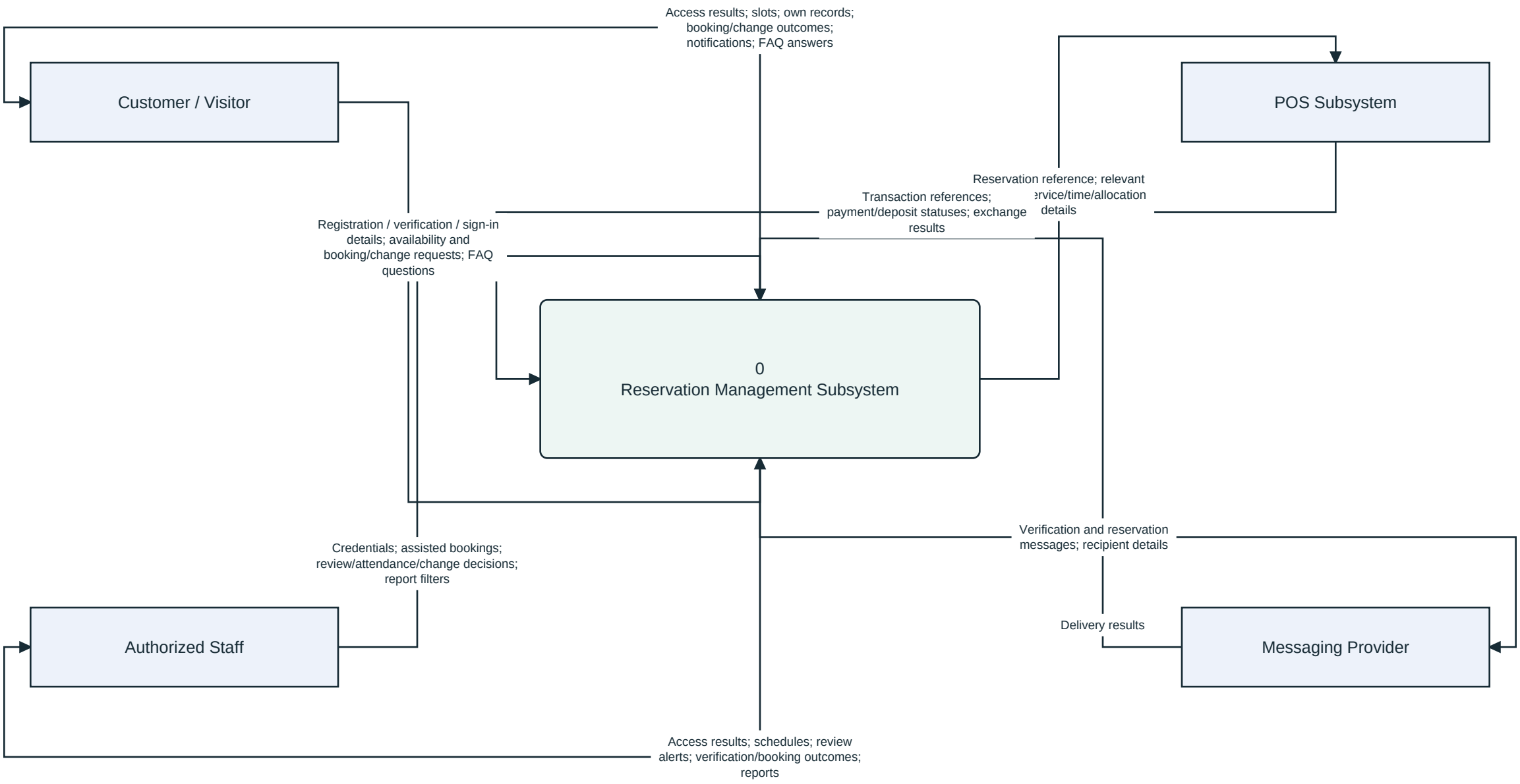
C. Attendance, changes and operational outputs



Change path is available for an existing reservation under approved permissions. Successful edits preserve lifecycle rules; changed terms may require staff reconfirmation.

DFD Context Diagram

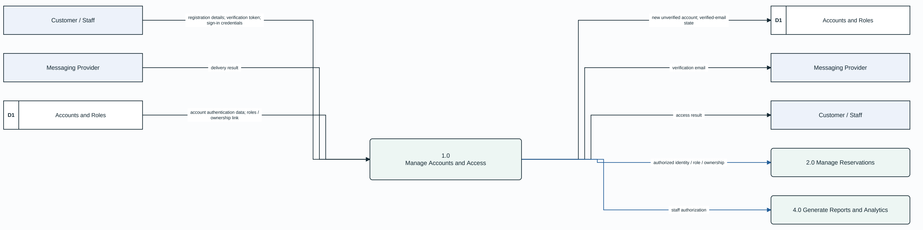
System boundary | arrows carry data, not decisions or execution order



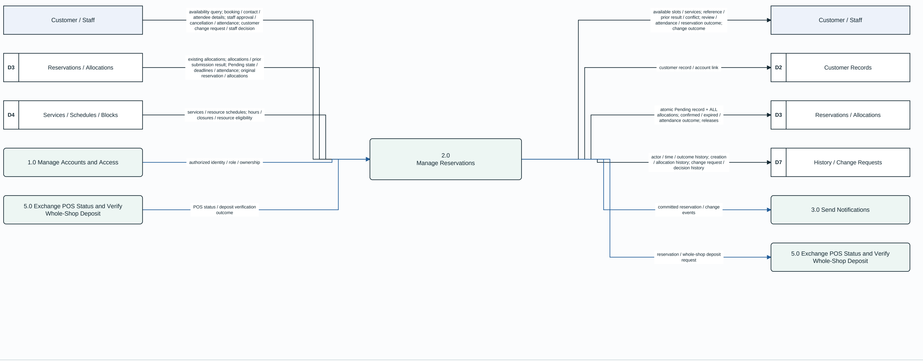
Level 1 DFD | Parent Process Boundary Reference

Six parent processes | repeated IDs identify the same entity, store or process | each panel shows one parent boundary

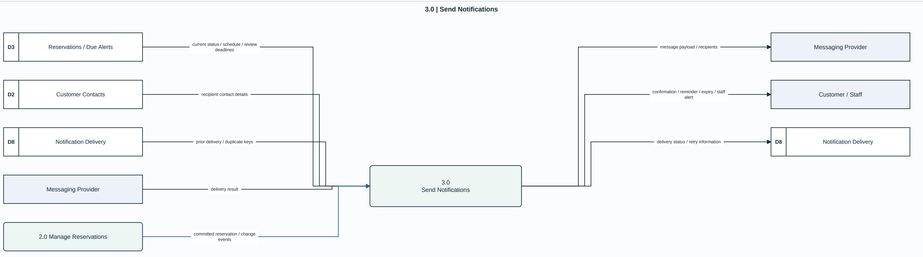
1.0 | Manage Accounts and Access



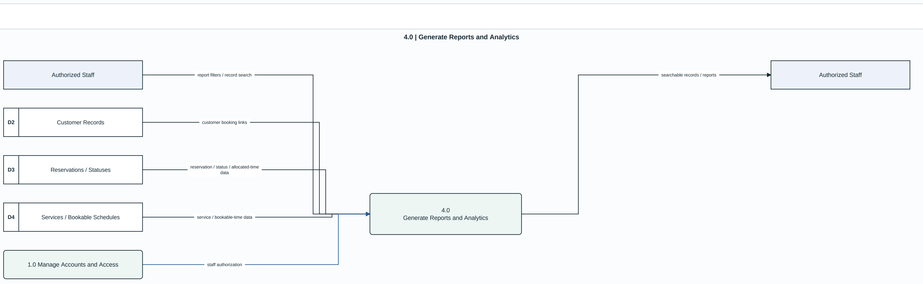
2.0 | Manage Reservations



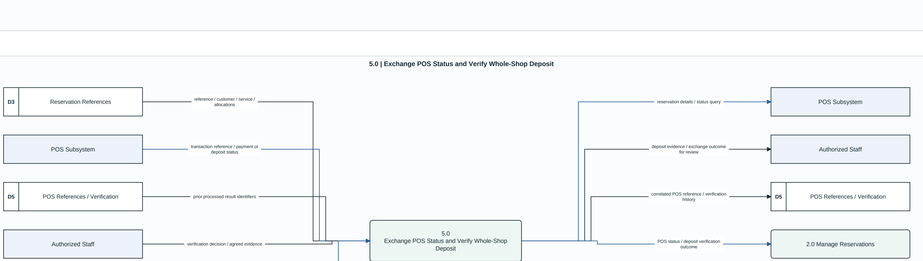
3.0 | Send Notifications



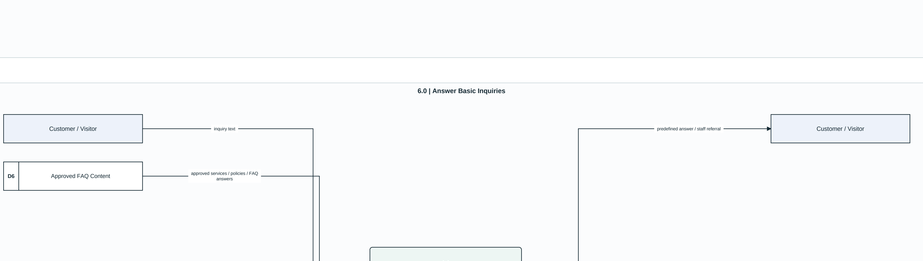
4.0 | Generate Reports and Analytics



5.0 | Exchange POS Status and Verify Whole-Shop Deposit

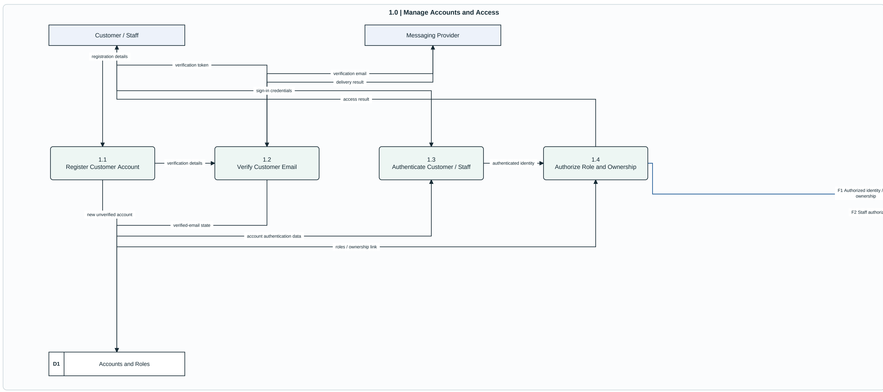


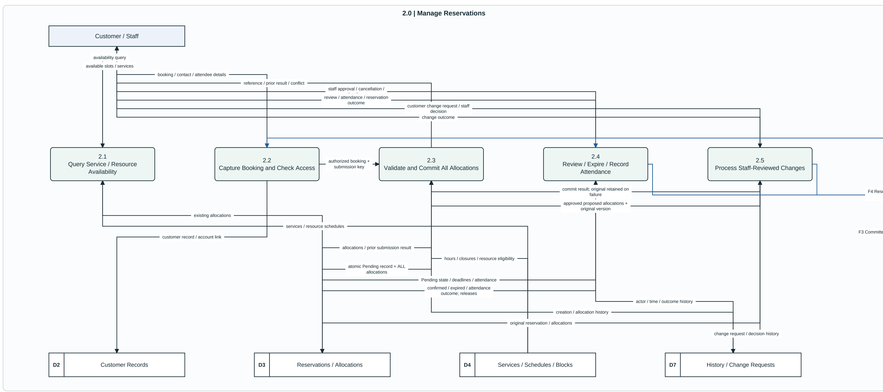
6.0 | Answer Basic Inquiries

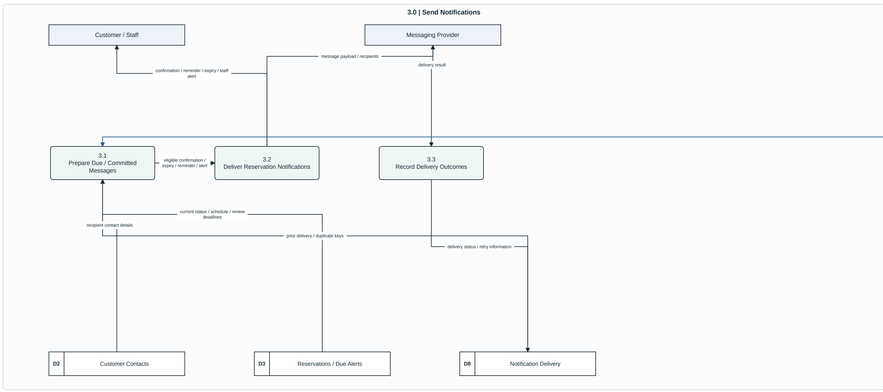


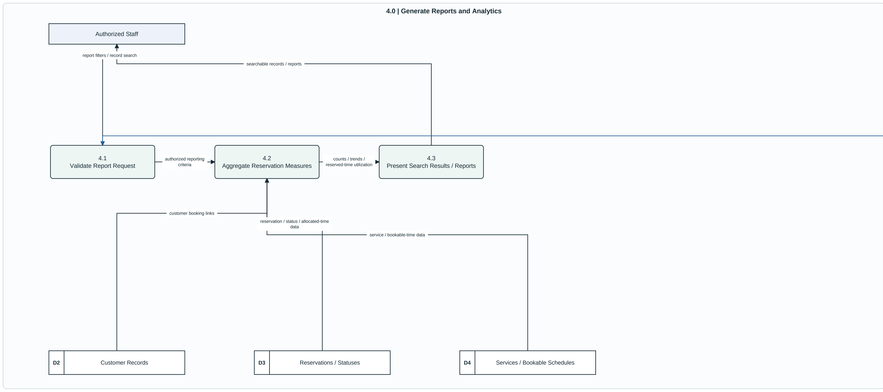
Level 2 DFD | Combined Decomposition of Processes 1.0-6.0

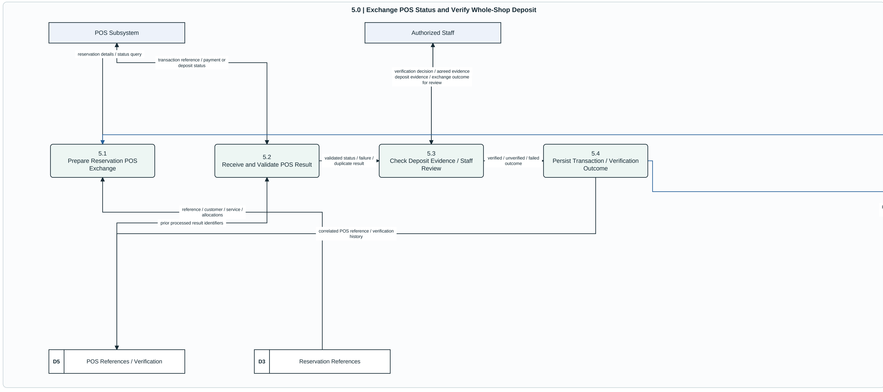
Data movement, not execution order | repeated store/entity IDs denote the same logical endpoint | blue: inter-process exchange







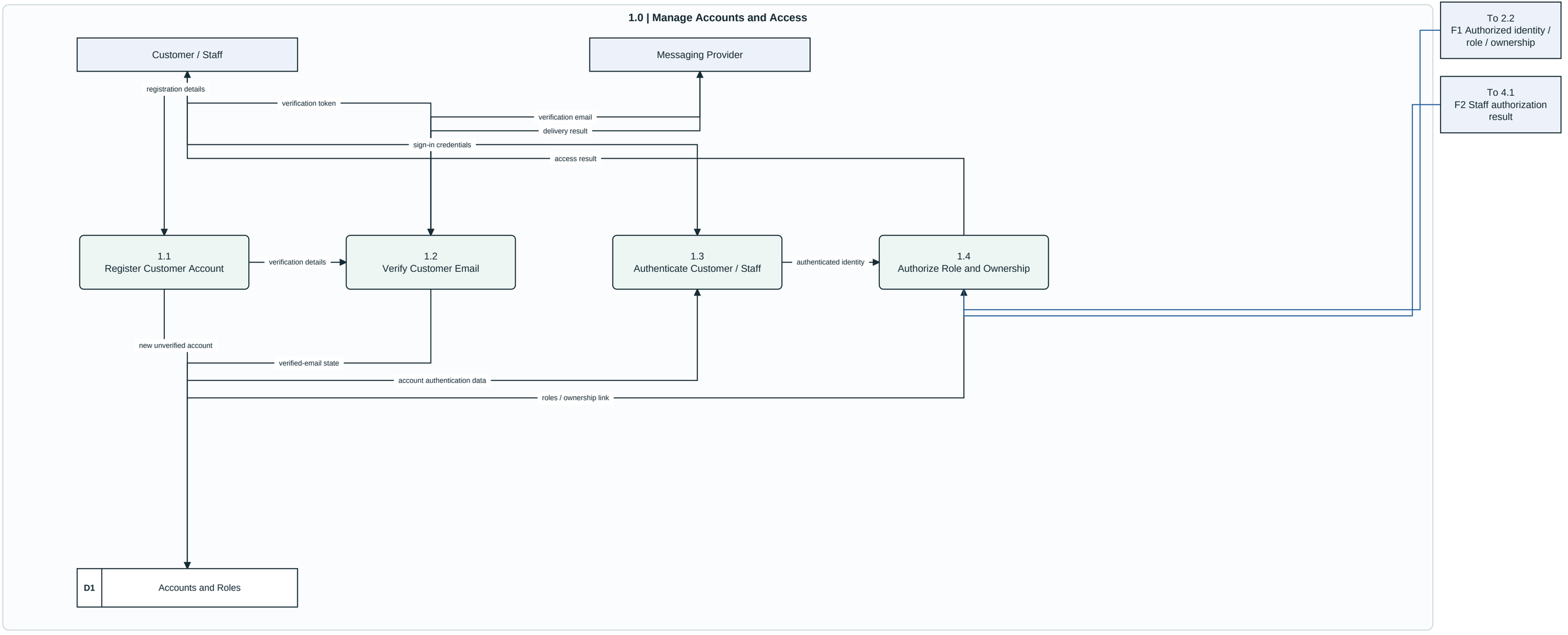






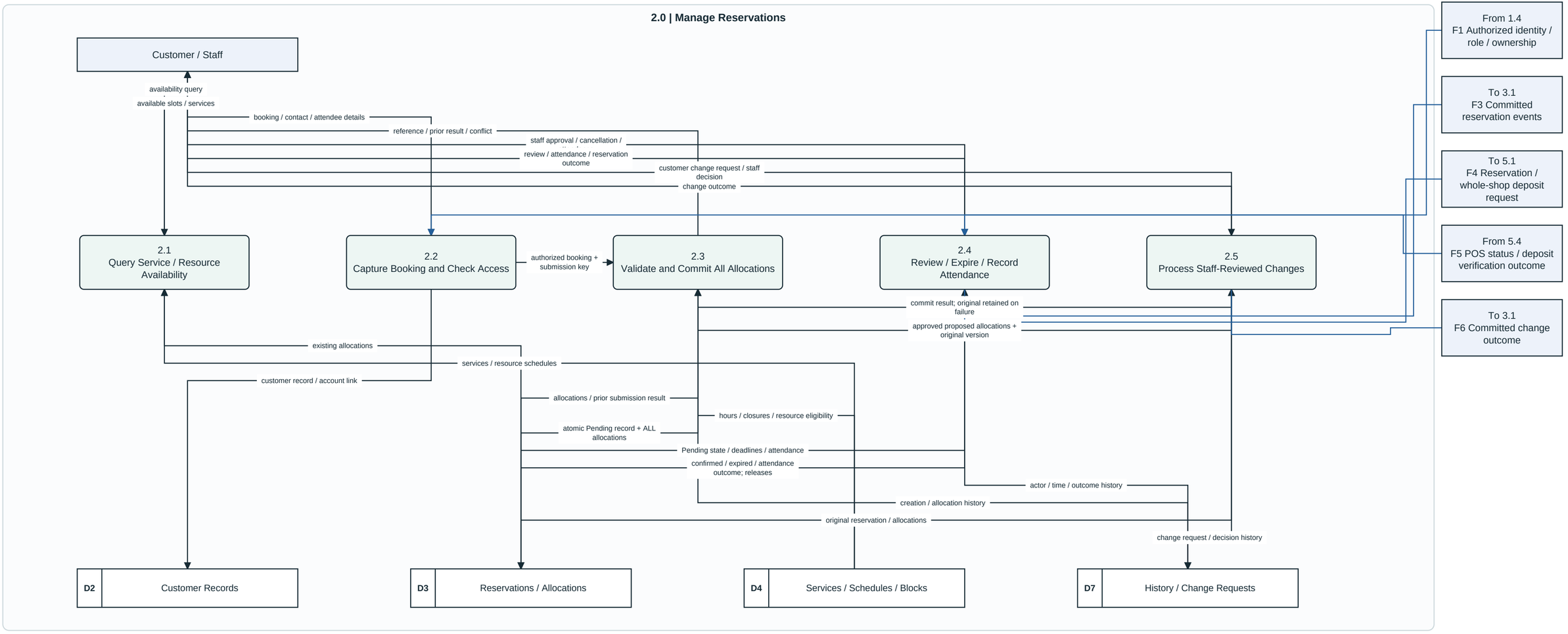
Level 2 DFD | 1.0 Manage Accounts and Access

Same process IDs and boundary flows as the combined DFD | see README for process rules and policies



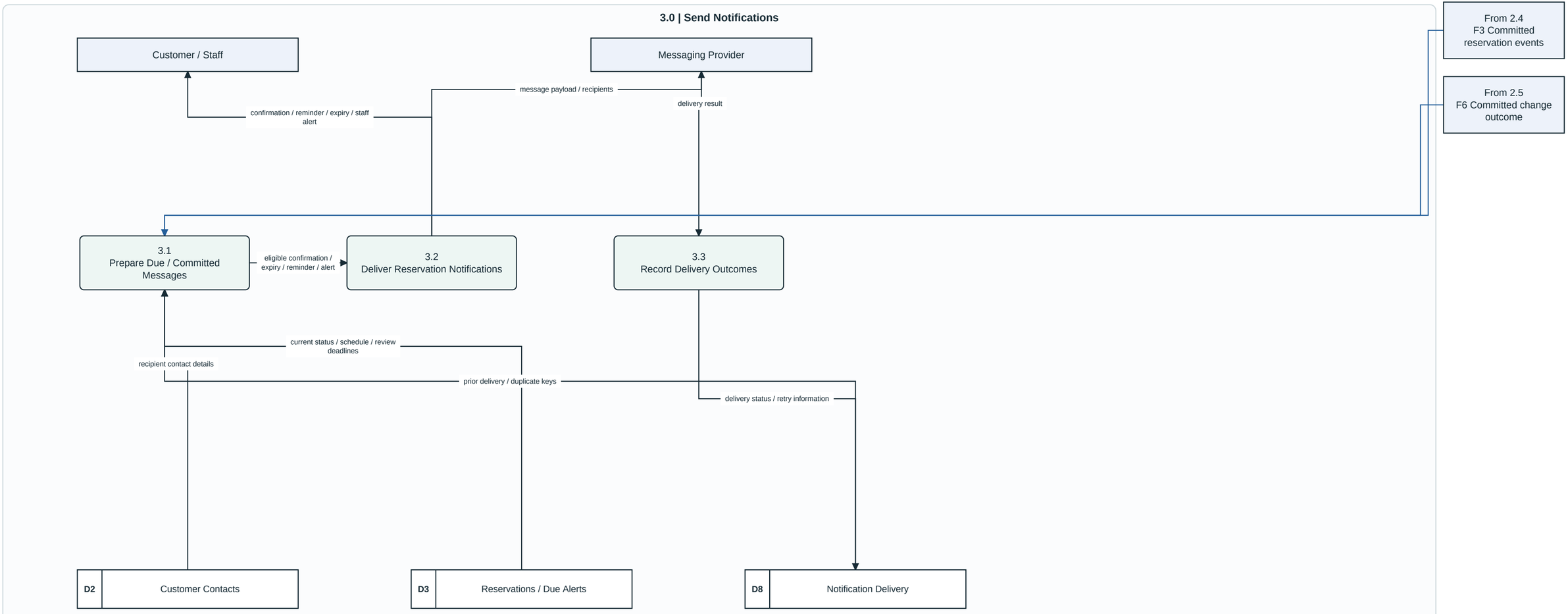
Level 2 DFD | 2.0 Manage Reservations

Same process IDs and boundary flows as the combined DFD | see README for process rules and policies



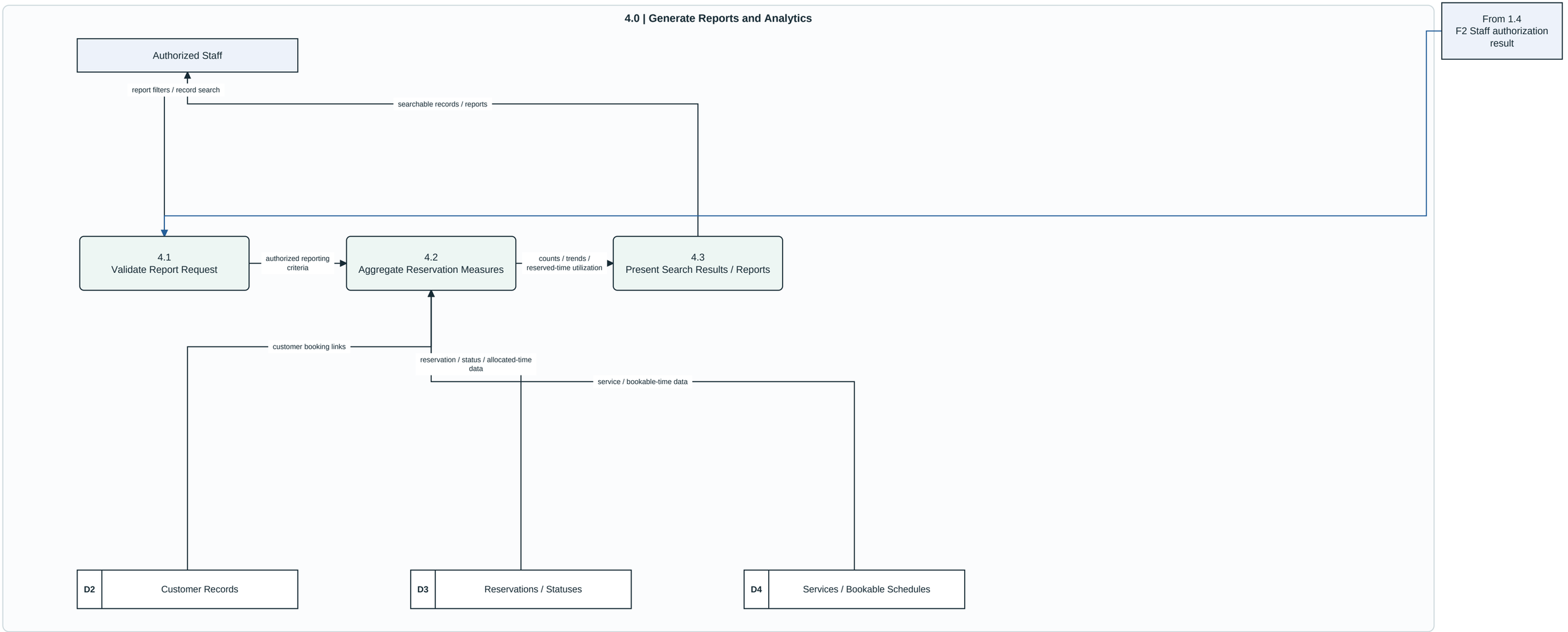
Level 2 DFD | 3.0 Send Notifications

Same process IDs and boundary flows as the combined DFD | see README for process rules and policies



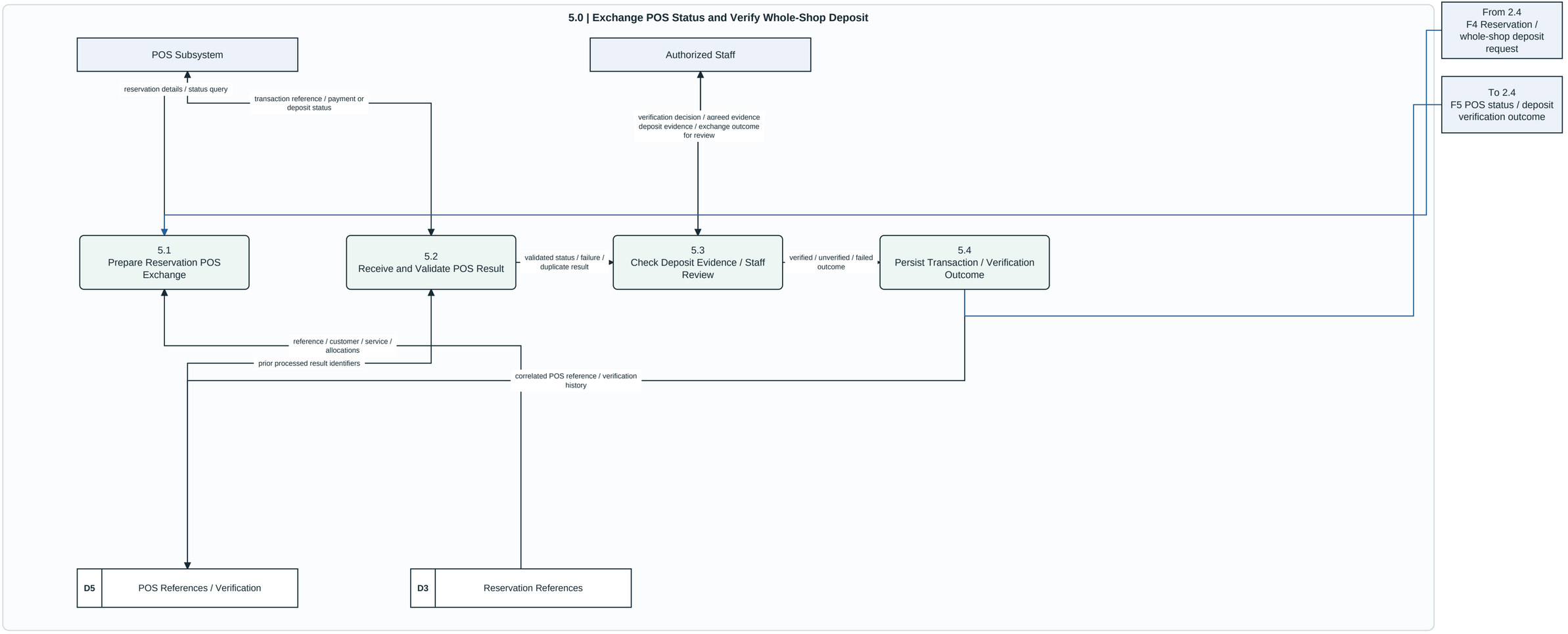
Level 2 DFD | 4.0 Generate Reports and Analytics

Same process IDs and boundary flows as the combined DFD | see README for process rules and policies



Level 2 DFD | 5.0 Exchange POS Status and Verify Whole-Shop Deposit

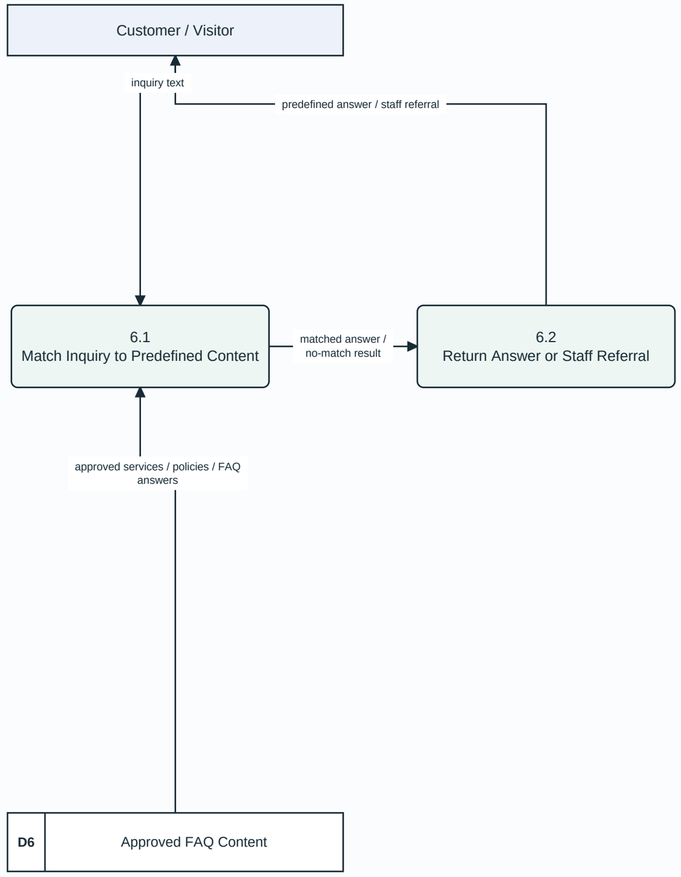
Same process IDs and boundary flows as the combined DFD | see README for process rules and policies



Level 2 DFD | 6.0 Answer Basic Inquiries

Same process IDs and boundary flows as the combined DFD | see README for process rules and policies

6.0 | Answer Basic Inquiries



Integration Flow Diagram | Reservation and POS

Logical sequence; transport/authentication/retries to be agreed | payment processing remains in POS

